SVI-JW: One Hyperbola, Two Languages

Raw geometry, trader coordinates, and honest arbitrage control

Vol-Fitter Technical Notes, No. 02 — alternative draft

Vol-Fitter

July 17, 2026

Abstract

SVI draws one expiry's price-implied total variance as a tilted hyperbola. That single picture explains both its durability and its limitations: the curve is smooth, strictly convex in log-moneyness, linear in both wings, cheap to fit, and unable to turn twice. The raw coefficients are excellent computational coordinates but poor market language. SVI-JW reads the same curve through five functionals instead: ATM variance, the ATM slope of total volatility, two normalized wing coefficients, and minimum variance. We build the raw formula from a symmetric toy, derive its geometry, define the JW handles in their natural units, and prove the full image of the coordinate map. Away from the ATM-at-the-minimum stratum the inverse is unique; on that stratum the five handles omit curvature and infinitely many raw slices share them. The inverse denominator is quadratic in the distance to this singular set, which explains its poor conditioning. We then separate four statements that are often conflated: convexity of total variance, positive/Lee-clean wings, non-negative Black density, and calendar ordering. In particular, the two production screens are not a butterfly certificate: the Axel Vogt slice has positive minimum variance and Lee coefficient 0.174, yet reaches $g = -0.033$. Finally we derive the unconstrained calibration map and analytic Jacobian, trace the optional residual blocks, and run a production JW→raw→fit→JW laboratory round trip with 5.6e-13 volatility-basis-point maximum error. Eight generated figures make the geometry, singularity, arbitrage gap, calibration check, speed result, and one-hyperbola rigidity visible rather than merely asserted.

Contents

1 One curve, two languages

A desk rarely asks for "more b" or "less m." It asks where ATM volatility is, how steeply the smile leaves ATM, how heavy the put wing is, and where the smile bottoms. An optimizer has almost the opposite preference: it wants a formula that evaluates in one pass, differentiates without numerical noise, and behaves sensibly outside the quoted strikes. Raw SVI and SVI-JW are the two languages that serve those two jobs.

The underlying object is not implied volatility itself. Fix an expiry date T, let F_T be its forward, and write

$$k = \log \frac{K}{F_T}, \qquad w(k) = \text{Black implied total variance at } k. \tag{1}$$

The normalized Black price depends on k and w, so w is recovered from a price without first choosing an annualization clock. Vol-Fitter keeps two clocks distinct:

$$I_t(k) = \sqrt{\frac{w(k)}{t}}, \qquad I_\tau(k) = \sqrt{\frac{w(k)}{\tau}}. \tag{2}$$

Here t is calendar year fraction and I_t is the ordinary market IV; τ is the event-weighted variance time of Note 11 and I_τ is the working IV used by the fit and display paths when that clock is enabled. At fixed price, changing the clock changes the annualized number, not w. When events are disabled, $\tau = t$. Unless qualified, I below means the working quantity I_τ.

Raw SVI postulates

$$\boxed{w(k) = a + b\left\{\rho(k-m) + \sqrt{(k-m)^2 + s^2}\right\}.} \tag{3}$$

The width is denoted by s so that σ remains available for implied volatility; the production dataclass calls this field `sigma`. The standard raw tuple is (a, b, ρ, m, s) with

$$b > 0, qquad|\rho| < 1, qquads > 0. \tag{4}$$

Raw SVI was introduced by Gatheral and became a market standard because its wings are linear in $|k|$ and the formula is unusually tractable [1, 2].

Invariant

1. Every finite optimizer vector maps to a smooth raw hyperbola satisfying (4); positivity and no-arbitrage require additional work.
2. The JW handles are exact functionals of that same w, not a second model. Their regular inverse is exact only on the domain proved in theorem 1.
3. The coded minimum and Lee rows are configurable soft screens. They do not certify non-negative density, and the calendar rows do not certify global cross-expiry order.
4. Production fits and stores raw SVI. JW is presently an analytical and test coordinate system: there is no runtime five-handle entry, bump, or export workflow.

The order of the note follows the mathematics rather than the software menu. We first understand the hyperbola, then learn to read it in JW coordinates, then ask when it is a legitimate option smile, and only then fit it.

2 Building the raw hyperbola

Start with the symmetric toy

$$w(k) = a + b\sqrt{k^2 + s^2}. \tag{5}$$

The constant a lifts the curve, b sets the far-wing steepness, and s rounds the corner. Replacing k by $k - m$ translates the rounded core. Adding $b\rho(k - m)$ tilts the two wings in opposite directions. This gives (3). The construction is simple, but it already warns us that m is merely the centre of the square root: after the tilt, it is not the minimum.

Put

$$x = k - m, qquad r = \sqrt{x^2 + s^2}, \qquad q = \sqrt{1 - \rho^2}. \tag{6}$$

These are the only local abbreviations needed for the raw geometry.

Proposition 1 (Geometry of one raw slice)**.** *Under (4), w is strictly convex and has one minimum:*

$$w'(k) = b\left(\rho + \frac{x}{r}\right), \qquad w''(k) = \frac{bs^2}{r^3} > 0, \tag{7}$$

$$k_\star = m - \frac{s\rho}{q}, \qquad w_\star = a + bsq. \tag{8}$$

Its positive left-wing coefficient and right-wing slope are

$$\beta_L = b(1 - \rho), \qquad \beta_R = b(1 + \rho), \tag{9}$$

in the sense that

$$w(k) = \beta_L|k| + O(1) \quad (k \to -\infty), \qquad w(k) = \beta_R k + O(1) \quad (k \to +\infty). \tag{10}$$

The signed derivative on the left tends to $-\beta_L$; calling β_L a "left slope" always means its positive magnitude.

Proof. Differentiation gives (7). Since $b, s > 0$, the second derivative is positive for every finite k, so a stationary point is the unique global minimum. Solving $w' = 0$ gives $x = -s\rho/q$, hence $r = s/q$ and (8). Finally $r = |x| + o(1)$ in either wing, which yields (10). □

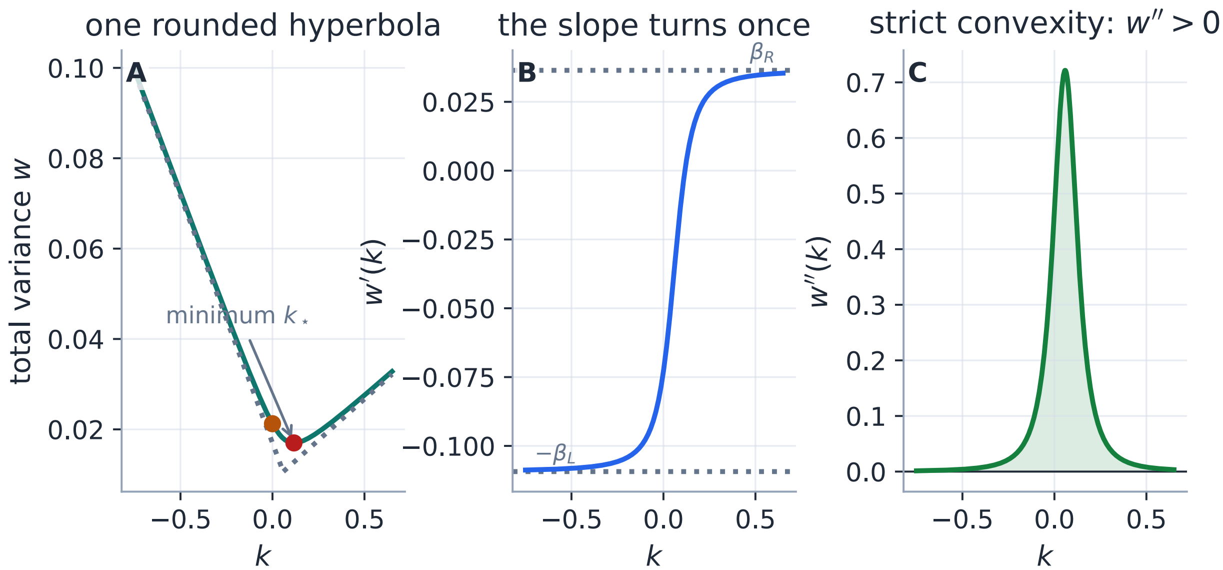

Figure 1: One hyperbola carries the entire model. The total-variance slice approaches two straight asymptotes (A), its derivative makes one smooth transition from $-\beta_L$ to β_R (B), and its curvature is positive everywhere (C). The last fact is a geometric statement about w; it is not yet a statement about convexity of option prices.

The raw parameters now have a precise, limited interpretation. Increasing a is a vertical translation. Increasing m translates the square-root core. But b, ρ, and s are coupled: b changes the minimum as well as both wings, ρ changes the minimum location as well as the two wing coefficients, and s changes both the depth and the breadth of the core.

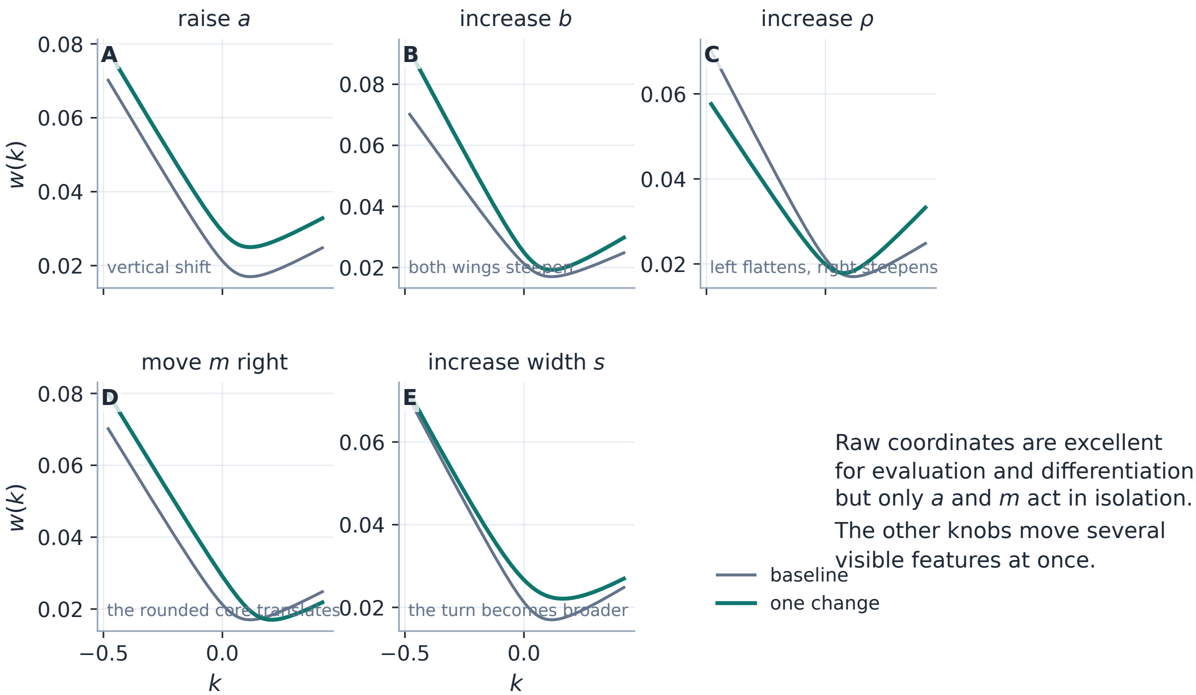

Figure 2: Raw parameters are computational coordinates, not independent market observables. Only a is a pure vertical move and only m is a pure translation of the square-root core. The other perturbations visibly alter several features at once, which is the motivation for a second coordinate chart.

Heuristic

Think of raw SVI as a rounded hinge. Far away, the hinge has forgotten its rounding and exposes two straight rays. Near the centre, s decides how quickly one ray turns into the other. The tilt ρ redistributes a fixed overall steepness between the put and call sides. This picture is more useful than memorizing five coefficient descriptions, but it still does not tell a trader the ATM level or ATM skew directly.

3 Reading the smile: the five JW handles

The clean way to introduce SVI-JW is to forget the raw coefficients for a moment and measure the curve itself. Let

$$w_0 = w(0) > 0, qquad w_\star = \min_k w(k). \tag{11}$$

Definition 1 (SVI-JW functionals)**.** At variance clock $\tau > 0$, the jump-wing coordinates of a smooth raw slice are

$$v = \frac{w_0}{\tau}, \qquad \psi = \left.\frac{\mathrm{d}\sqrt{w(k)}}{\mathrm{d}k}\right|_{k=0}, \qquad p = \frac{\beta_L}{\sqrt{w_0}}, \qquad c = \frac{\beta_R}{\sqrt{w_0}}, \qquad \tilde{v} = \frac{w_\star}{\tau}. \tag{12}$$

This definition separates intrinsic shape from annualization. The middle three handles ψ, p, c depend only on the price-derived w and are unchanged if the clock changes. Only the annualized levels v and $\widetilde{v}$ rescale. Production uses τ; an external calendar-market convention can replace it by t without changing the underlying slice.

The displayed meanings follow immediately:

$$I(0) = \sqrt{v}, qquad I'(0) = \frac{\psi}{\sqrt{\tau}}, qquad \beta_L = p\sqrt{v\tau}, \qquad \beta_R = c\sqrt{v\tau}, qquad I(k_\star) = \sqrt{\widetilde{v}} \quad \text{if } \widetilde{v} \geq 0. \tag{13}$$

Thus ψ is the ATM slope of *total volatility* $\sqrt{w}$, or equivalently $\sqrt{\tau}$ times the working-IV slope. The wing handles are normalized total-variance coefficients, not slopes of the IV plot.

Substituting (3) into the functional definition gives the classical raw-to-JW formulas [2, Sec. 3.3]:

$$\begin{aligned}
v &= \frac{a + b\{-\rho m + \sqrt{m^2 + s^2}\}}{\tau}, \\
\psi &= \frac{b}{2\sqrt{w_0}} \left(\rho - \frac{m}{\sqrt{m^2 + s^2}}\right), \\
p &= \frac{b(1-\rho)}{\sqrt{w_0}}, \qquad c = \frac{b(1+\rho)}{\sqrt{w_0}}, \\
\widetilde{v} &= \frac{a + bs\sqrt{1-\rho^2}}{\tau}.
\end{aligned} \tag{14}$$

Handle	Functional definition	What is visible on a chart	Common mistake
v	$w(0)/\tau$	ATM IV is $\sqrt{v}$	treating v as volatility
ψ	$(\sqrt{w})'(0)$	IV tangent is $\psi/\sqrt{\tau}$	calling ψ the plotted slope
p	$\beta_L/\sqrt{w_0}$	put coefficient is $p\sqrt{v\tau}$	reading it on the IV axis
c	$\beta_R/\sqrt{w_0}$	call coefficient is $c\sqrt{v\tau}$	reading it on the IV axis
$\widetilde{v}$	$w_\star/\tau$	minimum IV is $\sqrt{\widetilde{v}}$ if non-negative	assuming positivity from the name

Table 1: The five handles use three related but distinct languages: annualized variance, total-volatility skew, and normalized total-variance wings. Keeping those units explicit prevents nearly every common JW misreading.

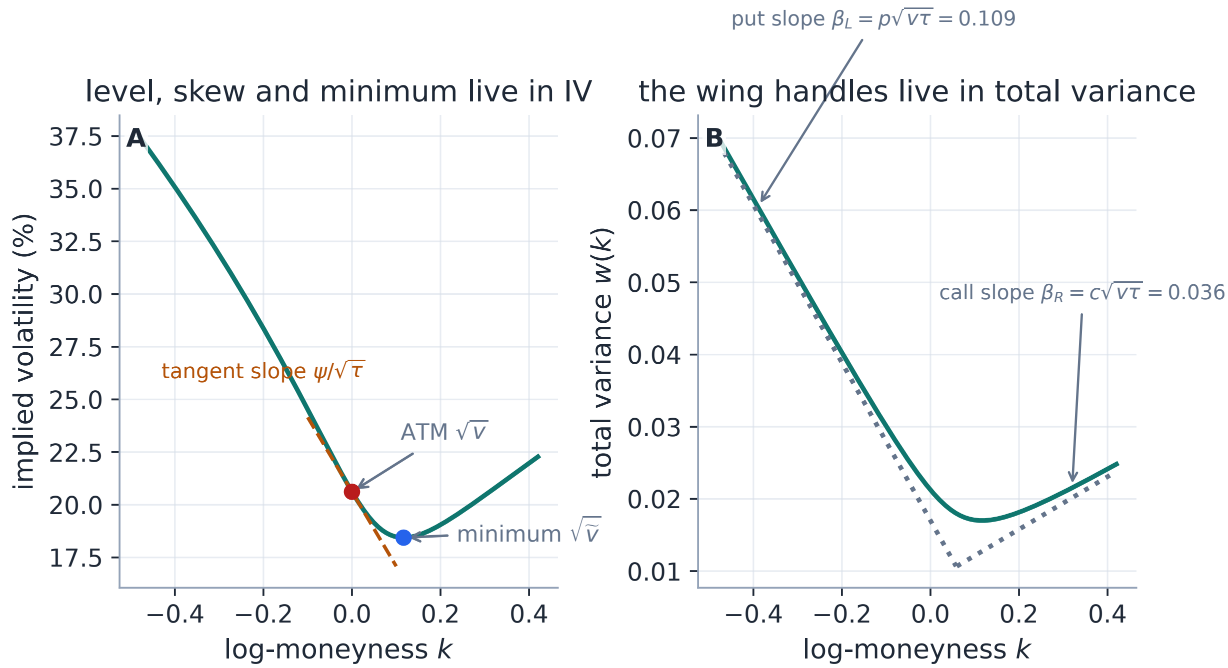

Figure 3: The five JW handles live on two charts. ATM level, ATM tangent, and minimum are naturally read in working IV (A). The put and call handles belong to the asymptotic total-variance rays (B). Plotting all five on one IV panel would give the wing handles units they do not have.

The name "jump-wing" is historical trader vocabulary; the five numbers are not parameters of a jump process. The parametrization, inspired by a Tim Klassen coordinate set, was documented alongside the other SVI forms by Gatheral and Jacquier [2]. Its value is descriptive: it turns one opaque coefficient vector into a level, a local slope, two tails, and a floor.

4 Changing coordinates, and where the chart tears

Reading JW from raw is just evaluation. Going the other way is more interesting because the five measurements must reconstruct a whole hyperbola. Let

$$w_0 = v\tau, qquad \chi = \frac{m}{\sqrt{m^2 + s^2}}. \tag{15}$$

The wings first determine the scale and tilt:

$$b = \frac{\sqrt{w_0}}{2}(p + c), qquad \rho = \frac{c - p}{c + p}. \tag{16}$$

The ATM skew then determines the normalized displacement:

$$\chi = \rho - \frac{4\psi}{p + c}. \tag{17}$$

For $|\chi| < 1$, write

$$D(\rho, \chi) = \frac{1 - \rho\chi}{\sqrt{1 - \chi^2}} - \sqrt{1 - \rho^2}. \tag{18}$$

The remaining three raw coordinates are

$$s = \frac{(v - \widetilde{v})\tau}{bD(\rho, \chi)}, \qquad m = \frac{\chi s}{\sqrt{1 - \chi^2}}, \qquad a = \widetilde{v}\tau - bs\sqrt{1 - \rho^2}. \tag{19}$$

The derivation is only the difference between ATM and minimum variance. Indeed, $\sqrt{m^2 + s^2} = s/\sqrt{1 - \chi^2}$, so subtracting $w_\star = \widetilde{v}\tau$ from $w_0 = v\tau$ produces bsD.

The factor D has a useful geometric form. Set $\rho = \sin\alpha$ and $\chi = \sin\gamma$, with both angles in $(-\pi/2, \pi/2)$. Then

$$D(\rho, \chi) = \frac{1 - \cos(\alpha - \gamma)}{\cos\gamma} \geq 0, \tag{20}$$

and equality holds exactly when $\chi = \rho$. This one observation supplies the domain and the singularity.

Theorem 1 (The full smooth, nondegenerate JW image)**.** *Fix $\tau > 0$ and restrict raw SVI to $b > 0$, $|\rho| < 1$, $s > 0$, and $w_0 > 0$.*

1. *A JW point has a unique regular inverse if and only if*

$$v > 0, qquadp > 0, quadc > 0, qquad -\frac{p}{2} < \psi < \frac{c}{2}, qquad\psi \neq 0, qquad\widetilde{v} < v. \tag{21}$$

   On this set, (16)–(19) *return that inverse.*

2. *The image also contains the singular stratum*

$$v > 0, qquadp > 0, quadc > 0, qquad\psi = 0, qquad\widetilde{v} = v. \tag{22}$$

   Every point on this stratum is represented by infinitely many raw slices. There are no other smooth, nondegenerate points in the image.

Proof. If $p, c > 0$, (16) gives $b > 0$ and $\rho \in (-1, 1)$. Conversely, a smooth raw slice with $b > 0$ and $|\rho| < 1$ has $p, c > 0$. From (17),

$$\begin{aligned} |\chi| < 1 &\iff \rho - 1 < \frac{4\psi}{p + c} < \rho + 1 \\ &\iff -\frac{p}{2} < \psi < \frac{c}{2}, \end{aligned}$$

where $(\rho - 1)(p + c) = -2p$ and $(\rho + 1)(p + c) = 2c$. This is exactly the condition for finite m/s.

By (20), $D > 0$ precisely when $\chi \neq \rho$, which by (17) is $\psi \neq 0$. The first equation in (19) then gives $s > 0$ precisely when $\widetilde{v} < v$. All remaining coordinates are unique algebraic consequences, proving the regular statement and its converse.

If $\psi = 0$, then $\chi = \rho$. On a raw slice this is equivalent to $k_\star = 0$ by (8); hence ATM is the minimum and $\widetilde{v} = v$. Conversely, fix any v, p, c satisfying (22). Equations (16) determine b, ρ. For *any* $s > 0$, set

$$m = \frac{\rho s}{\sqrt{1 - \rho^2}}, qquada = v\tau - bs\sqrt{1 - \rho^2}. \tag{23}$$

Then $k_\star = 0$, $w_\star = w_0 = v\tau$, and all five JW functionals agree. The free width proves non-identification. The preceding necessity arguments exclude every other smooth nondegenerate case. $\square$

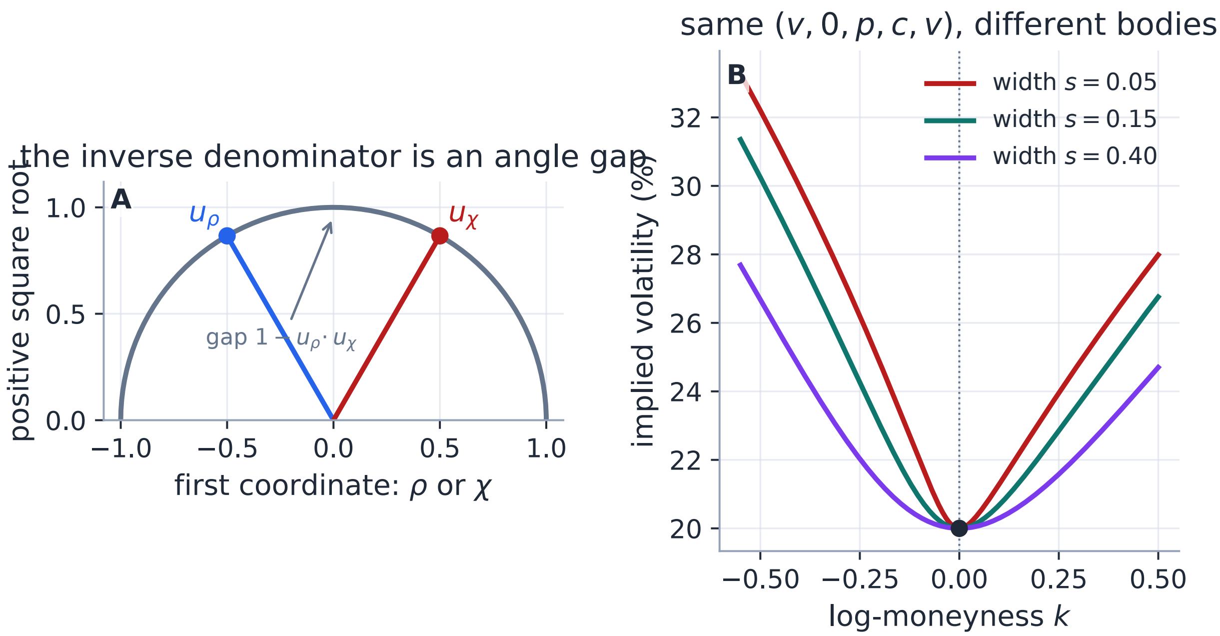

Figure 4: The inverse denominator measures an angular separation on the unit semicircle (A). When the two unit vectors coincide, $\psi = 0$ and the chart loses width information. Three visibly different raw bodies can then share the identical tuple $(v, 0, p, c, v)$ (B): JW fixes ATM and both asymptotic rays but omits curvature at the common minimum.

4.1 Boundaries and conditioning

The theorem distinguishes *representability* from *regular invertibility*. The most useful boundary cases are collected here.

JW boundary	What happens
$\psi = -p/2$ or $\psi = c/2$	$\lvert\chi\rvert = 1$; no finite smooth displacement m/s
$\psi = 0$, $\tilde{v} = v$	representable, but infinitely many raw widths
$\psi = 0$, $\tilde{v} < v$	no finite smooth raw slice
$\psi \neq 0$, $\tilde{v} = v$	$s = 0$, the nonsmooth V-shaped boundary
$p = 0$ or $c = 0$	$\lvert\rho\rvert = 1$, a one-wing-degenerate boundary
$p = c = 0$	$b = 0$, the constant-slice boundary
$v = 0$	the normalization by $\sqrt{w_0}$ is undefined
$\tilde{v} < 0$ inside the regular domain	algebraically invertible, financially unusable

Table 2: The regular domain is open for a reason. Equality cases are not one generic failure mode; they lead to distinct degeneracies.

Near $\chi = \rho$, the angular gap is quadratic:

$$D(\rho, \chi) \sim \frac{(\chi - \rho)^2}{2(1 - \rho^2)^{3/2}}, \qquad \chi \to \rho. \tag{24}$$

Because $\chi - \rho = -4\psi/(p + c)$, the denominator is $O(\psi^2)$. Holding $v - \widetilde{v} > 0$ fixed therefore sends s to infinity as $\psi \to 0$; a finite approach to the singular family requires $v - \widetilde{v} = O(\psi^2)$. The limiting ratio is precisely the curvature information missing from the five handles.

The form (18) also subtracts nearly equal numbers. An exactly equivalent, more stable evaluation is

$$D(\rho, \chi) = \frac{(\rho - \chi)^2 + \left(\sqrt{1 - \rho^2} - \sqrt{1 - \chi^2}\right)^2}{2\sqrt{1 - \chi^2}}, \tag{25}$$

with the square-root difference rationalized in code. This removes avoidable cancellation but cannot restore a coordinate that the handles do not identify.

Caution

The production `jw_to_raw` function implements the direct formulas without domain guards and has no runtime API/UI caller; tests and document generators call it with controlled inputs. Inputs strictly outside the skew band typically generate NaNs, skew-band endpoints create zero denominators, and $\widetilde{v} \geq v$ returns a degenerate or negative width. Any future user-facing JW workflow should validate (21), treat (22) separately, and use a cancellation-resistant denominator such as (25).

5 Convex total variance is not an arbitrage certificate

The phrase "convex smile" hides two different derivatives. Raw SVI has $w''(k) > 0$ by construction. A butterfly spread, however, tests the second derivative of an *option price with respect to strike*. Those statements are not equivalent.

It helps to keep four levels separate.

Level	Mathematical statement	Production status
Raw structure	$b > 0$, $\lvert\rho\rvert < 1$, $s > 0$	hard reparametrization for finite latent coordinates
Positive/Lee interior	$w_\star > 0$, $\beta_L < 2$, $\beta_R < 2$	weak closure screened softly: $w_\star \geq 0$, $\max(\beta_L, \beta_R) \leq 2$
Butterfly clean	non-negative Black density and correct call boundary	measured; not certified by the two core rows
Calendar clean	$w_{T_1}(k) \leq w_{T_2}(k)$ for every k when $T_1 < T_2$	finite-weight rows on selected grids; not structural

Table 3: Four increasingly strong meanings of "valid." The first is hard, the second is softly screened, and the last two require conditions beyond a single raw coefficient box.

5.1 The exact one-slice criterion

For normalized strike $y = e^k$, let $B(k, w(k))$ be the normalized Black call and set

$$d_\pm(k) = -\frac{k}{\sqrt{w(k)}} \pm \frac{\sqrt{w(k)}}{2}. \tag{26}$$

Proposition 2 (Durrleman's density factor)**.** *Let $w \in C^2(\mathbb{R})$ be strictly positive. The strike density implied by the Black call is*

$$\frac{\partial^2 B}{\partial y^2} = \frac{\varphi(d_-(k))}{y\sqrt{w(k)}}\, g(k), \tag{27}$$

where

$$g(k) = \left(1 - \frac{k w'(k)}{2w(k)}\right)^2 - \frac{w'(k)^2}{4}\left(\frac{1}{w(k)} + \frac{1}{4}\right) + \frac{w''(k)}{2}. \tag{28}$$

Consequently density non-negativity is equivalent to $g(k) \geq 0$ for every k. A standard no-butterfly smile additionally needs the far-call boundary

$$\lim_{k\to+\infty} d_+(k) = -\infty, \tag{29}$$

equivalently the normalized call tends to zero at infinite strike.

Proof. Twice differentiating the Black call with the chain rule gives (27); the prefactor is strictly positive, so its sign is the sign of g. The boundary is separate because a non-negative local density need not have the correct total mass or first moment. This is the standard Durrleman/Gatheral–Jacquier criterion [4, 2]. □

For raw SVI, Lee's wing bound appears directly in g.

Proposition 3 (Raw-SVI tail limits)**.** *For a structurally valid positive raw slice,*

$$\lim_{k\to-\infty} g(k) = \frac{4-\beta_L^2}{16}, qquad \lim_{k\to+\infty} g(k) = \frac{4-\beta_R^2}{16}. \tag{30}$$

Moreover (29) holds exactly when $\beta_R < 2$; at $\beta_R = 2$, raw SVI has $d_+(k) \to 0$.

Proof. In either wing $w \sim \beta|k|$, $|w'| \to \beta$, and $w'' \to 0$. The first term of (28) tends to $1/4$ and the second to $\beta^2/16$, giving (30). On the right,

$$d_+(k) = \sqrt{k}\left(-\frac{1}{\sqrt{\beta_R}} + \frac{\sqrt{\beta_R}}{2}\right) + o(\sqrt{k}),$$

whose coefficient is negative exactly for $\beta_R < 2$. When $\beta_R = 2$, the leading terms cancel and the raw linear asymptotic gives the stated zero limit. □

The strict inequalities are the clean theoretical interior. The production hinges use the weak closure because numerical penalties need a boundary. Similarly, the theoretical positive-smile condition is $w_\star > 0$. A smooth nondegenerate raw slice touching zero at one finite strike is not a harmless edge case; it lies outside the butterfly-free raw-SVI domain [5]. The coded condition $w_\star \geq 0$ is a screen, not a theorem.

5.2 What the two coded screens look like in JW coordinates

Proposition 4 (Core screens in JW units)**.** *The minimum and Lee quantities satisfy*

$$w_\star = \widetilde{v}\tau, qquad \max(\beta_L, \beta_R) = \max(p, c)\sqrt{v\tau}. \tag{31}$$

Thus the production weak screens are $\widetilde{v} \geq 0$ *and* $\max(p, c)\sqrt{v\tau} \leq \beta_{\max}$*, with default* $\beta_{\max} = 2.0$*.*

Proof. The first identity is the definition of $\widetilde{v}$. The second follows from $p = \beta_L/\sqrt{w_0}$, $c = \beta_R/\sqrt{w_0}$, and $w_0 = v\tau$. □

The screens are necessary ingredients, not sufficient ones. The sharpest way to see the gap is a slice that passes both.

Case file: the slice that looks innocent

Axel Vogt's counterexample, reproduced as Example 3.1 by Gatheral and Jacquier [2], has

$$a = -0.0410, \quad b = 0.1331, \quad \rho = 0.3060, \quad m = 0.3586, \quad s = 0.4153.$$

Its minimum total variance is 0.0116, and its larger Lee coefficient is only 0.174. Both production core screens pass comfortably. Nevertheless g reaches -0.033 near $k = 0.88$, so the Black density is negative there. Strict convexity of w has not protected strike convexity of the call price.

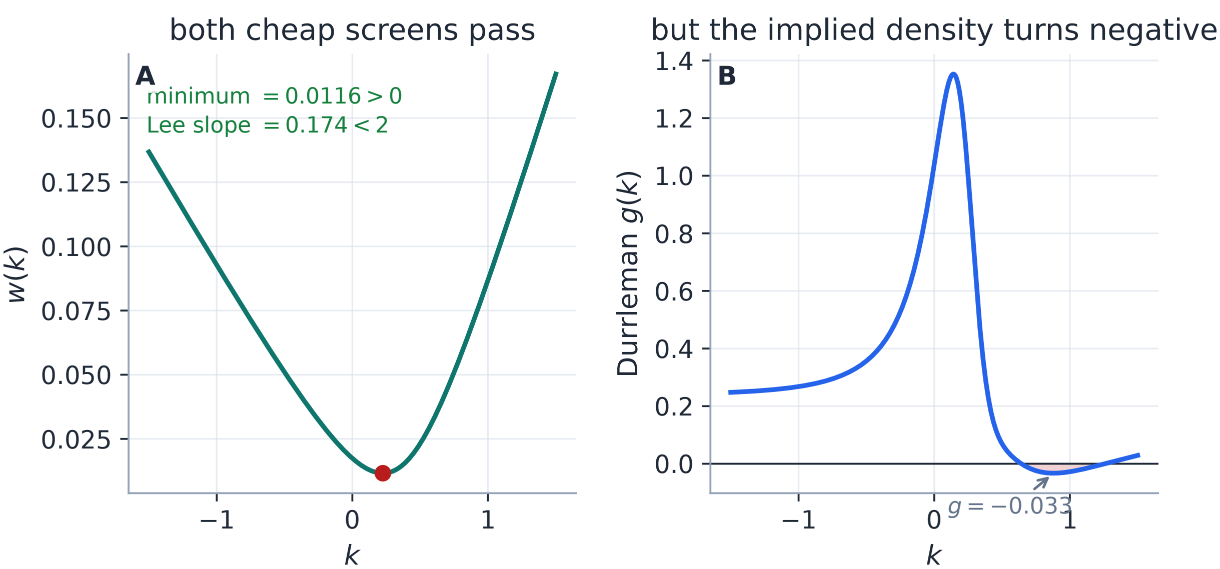

Figure 5: The counterexample is the note's main honesty check. Its positive, smooth, strictly convex total variance passes the minimum and Lee screens (A), while Durrleman's density factor becomes negative (B). The cheap screens control common failures; they do not characterize the butterfly-free domain.

Martini and Mingone give a full characterization of the no-butterfly raw-SVI domain [5]. It is exact but not a single short inequality: the practical test involves parameter rescaling, root finding, and numerical minimization. Vol-Fitter's current raw-SVI calibrator does not implement that certified domain. It instead bounds, measures, and optionally repairs through four separate layers.

1. **Core screens.** Every fit appends the two minimum/Lee hinge rows of (32) below.
2. **Optional extrapolation enforcement.** With `extrapEnforce` on, tapered rows penalize sampled $g < 0$, calendar crossings against the previous displayed slice, and inconsistent wing order in the extrapolated region. Only this small block is finite-differenced.
3. **Advisory diagnostics.** The Quality path reports remaining extrapolated-region violations using model-native quantities: closed-form raw SVI derivatives here, structural density for LQD, and the SIV diagnostic for that family.
4. **Publish-time projection.** The export path can apply the model-agnostic discrete wing projection of Note 09 while leaving the fitted core fixed.

No item in that list silently upgrades the raw fit to a global certificate. The exact criterion remains proposition 2; the production layers state what they sample or alter.

The two core residuals are

$$r_{\text{core}} = W \begin{pmatrix} [-w_\star]_+ \\ [\max(\beta_L, \beta_R) - \beta_{\max}]_+ \end{pmatrix}, \qquad [x]_+ = \max(x, 0), \tag{32}$$

with default residual multiplier $W = 1000$. Because least squares squares residuals, the objective coefficient is W^2. The two components do not even have the same natural unit: one is total variance and one is a wing coefficient. W is therefore an engineering multiplier, not a financial variance weight. At a strictly feasible point both rows and their Jacobian rows vanish, so the penalized and unpenalized objectives agree locally. That fact does *not* imply identical optimizer paths or bitwise identical fits for arbitrary data.

6 How production fits the slice

The optimizer never works directly with a bounded raw vector. It receives $\theta \in \mathbb{R}^5$ and maps

$$a = \theta_1, qquadb = \operatorname{softplus}(\theta_2), \qquad \rho = \tanh(\theta_3), \qquad m = \theta_4, qquads = e^{\theta_5}. \tag{33}$$

For every finite θ in exact arithmetic, $b > 0$, $|\rho| < 1$, and $s > 0$. This is the hard structural guarantee. It says nothing about $w_\star$, g, or calendar order.

6.1 One residual stack

Suppose price inversion has produced quote total variances w_i at log-moneyness k_i, with working IVs $I_i = \sqrt{w_i/\tau}$ and non-negative weights ω_i. In mid mode the data rows are

$$r_i(\theta) = \sqrt{\omega_i} \left(\sqrt{\frac{w(k_i;\theta)}{\tau}} - I_i \right). \tag{34}$$

The core problem is

$$\min_{\theta \in \mathbb{R}^5} \frac{1}{2} \left\| \begin{pmatrix} r_{\text{data}}(\theta) \\ r_{\text{core}}(\theta) \\ r_{\text{calendar}}(\theta) \\ r_{\text{optional}}(\theta) \end{pmatrix} \right\|_2^2 . \tag{35}$$

With no optional target the last two blocks are empty. Bid–ask and haircut modes replace the mid data rows by a vol-space band hinge plus a small mid anchor. The surfaced quote-weight choices are `equal` and `tv_density`; the low-level calibrator also accepts arbitrary weights, including vega-squared research weights.

The raw structure does not prevent a trial a from making w negative. To keep the IV residual evaluable, the calibration residual floors trial total variance at 10^{-12} inside the square root and sets the derivative to zero where the floor is active. The `RawSVI.implied_vol` method itself has no such floor; this is a solver device, not a model definition.

6.2 A geometric, deliberately rough start

The initializer sorts the quotes, uses the observed argmin as a seed for m, and estimates two finite-span slopes from the endpoints to the array midpoint. Those slopes seed b and ρ; the width starts at one tenth of the quoted k span, floored at 0.01, and a is backed out from the observed minimum with a small positive floor. Each transformation in (33) is then inverted to produce θ_0.

This is an initializer, not an estimator. The observed argmin is generally an estimate of $k_\star$, not of raw m; the array midpoint need not be ATM on an uneven or one-sided chain; and finite quoted slopes are not asymptotic wings. The construction is useful because it usually lands in the correct basin, not because its intermediate numbers have structural meaning.

6.3 The analytic Jacobian

The chain has three layers:

$$\theta \longmapsto (a, b, \rho, m, s) \longmapsto w(k) \longmapsto I(k) \longmapsto r_i.$$

With x and r from (6), the raw partial derivatives are

$$\frac{\partial w}{\partial(a, b, \rho, m, s)} = \left(1,\ \rho x + r,\ bx,\ -b\left(\rho + \frac{x}{r}\right),\ \frac{bs}{r}\right). \tag{36}$$

The latent-coordinate factors are

$$\frac{\mathrm{d}b}{\mathrm{d}\theta_2} = 1 - e^{-b}, \qquad \frac{\mathrm{d}\rho}{\mathrm{d}\theta_3} = 1 - \rho^2, \qquad \frac{\mathrm{d}s}{\mathrm{d}\theta_5} = s, \tag{37}$$

and the IV factor is

$$\frac{\partial I}{\partial w} = \frac{1}{2\tau I} = \frac{1}{2\sqrt{\tau w}}. \tag{38}$$

Multiplying these factors gives the data Jacobian without another model evaluation. Active hinge rows use the derivative of their linear branch and inactive rows use zero; at a kink the implementation chooses the inactive subgradient. In particular the $|\rho|$ term in the Lee row uses $\operatorname{sgn}(\rho)$, with the zero subgradient at $\rho = 0$.

The analytic module covers the mid or band data block, both core rows, and the sampled calendar floor. A var-swap target or persistence-prior block switches the whole fit to finite differences because those rows are not differentiated there. Extrapolation enforcement keeps a hybrid Jacobian: the dominant core remains analytic and only the small extrapolation block is centrally finite-differenced.

Residual block	Purpose	Jacobian path
mid or band data	fit working IV, or stay inside bid–ask/haircut band	analytic
minimum and Lee	weak core screens (32)	analytic subgradient
calendar floor	reduce sampled crossings against the nearer expiry	analytic subgradient
extrapolation	sampled g, calendar, and wing-order hinges	analytic core + FD block
var-swap quote	pull replicated fair variance toward a quote	finite-difference fit
strike/operator prior	preserve under-observed prior shape or operators	finite-difference fit

Table 4: Every optional target is expressed in the native currency of raw SVI, but not every row has yet been differentiated analytically. “Analytic” is a performance statement, not an arbitrage statement.

Performance

The solver remains Levenberg–Marquardt. A tested switch to trust-region reflective was slower on noisy real chains because it took more iterations through hinge changes; the useful change was the Jacobian, not the optimizer. On the generated 25-quote case, a warmed median-of-five run measures 3.06 ms with finite differences and 1.21 ms analytically, a 2.53× speed-up with objective costs agreeing to 2.7e-33. The separately recorded spike-regime measurement was 26.3→10.2 ms per real node (2.58×). The first number is machine-local and regenerated; the second is a dated historical benchmark recorded in `backend/backtest/FINDINGS_calibration_arb.md`.

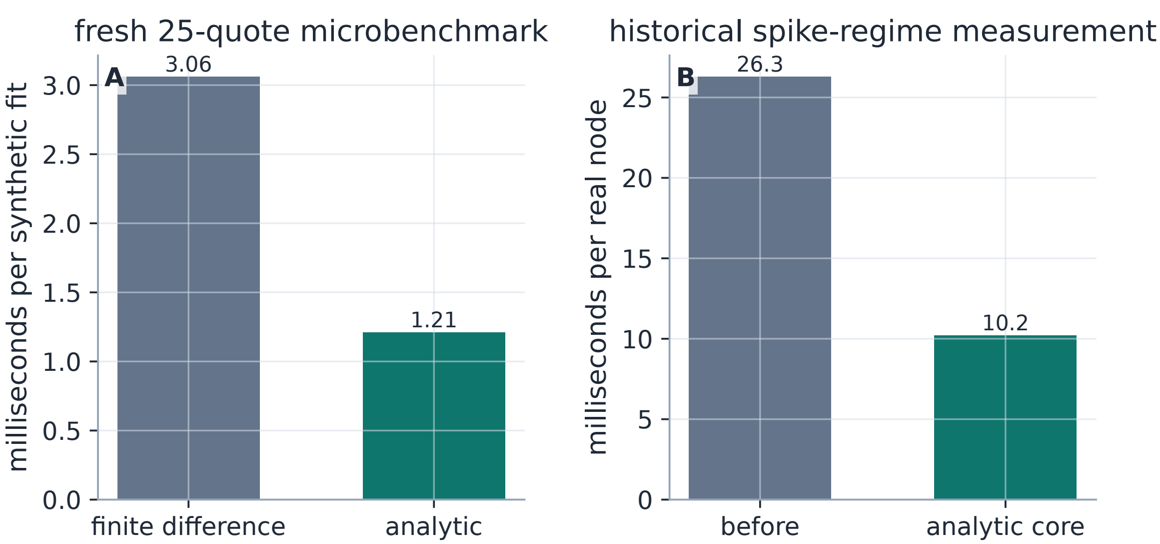

Figure 6: The analytic Jacobian removes repeated residual evaluations rather than changing the optimizer. The fresh synthetic timing (A) and the historical real-node timing (B) have different scales and provenance, but both isolate the same implementation change.

7 Laboratory case: JW to raw to fit to JW

An exact-family, noise-free recovery is not market evidence. It is the right laboratory test for a coordinate conversion and a Jacobian: any visible miss is an implementation error.

Take $\tau = 0.5$ and

$$(v, \psi, p, c, \widetilde{v}) = (0.0425, -0.25, 0.75, 0.25, 0.034). \tag{39}$$

The production `jw_to_raw` converter builds the target. We sample 25 noise-free total-variance quotes on $[-0.35, 0.30]$, fit production raw SVI from its geometric start, then read JW back using (14). This is a genuine

$$\text{JW} \longrightarrow \text{raw target} \longrightarrow \text{raw fit} \longrightarrow \text{JW readout}$$

round trip.

Table 5: Recovered raw coordinates.

Parameter	Value
a	+0.010625
b	+0.072887
ρ	-0.500000
m	+0.058310
s	+0.100995

Table 6: Recovered JW functionals.

Handle	Value
v	+0.042500
ψ	-0.250000
p	+0.750000
c	+0.250000
$\widetilde{v}$	+0.034000

The maximum quote error is 5.6e-13 vol bp after 21 residual evaluations. The maximum absolute JW round-trip error is 3.9e-16, and the recovered wing coefficients are 0.1093 and 0.0364. On the plotted diagnostic grid, g stays above 0.343. These numbers test algebra and implementation; they do not measure SVI's ability to fit a non-SVI market.

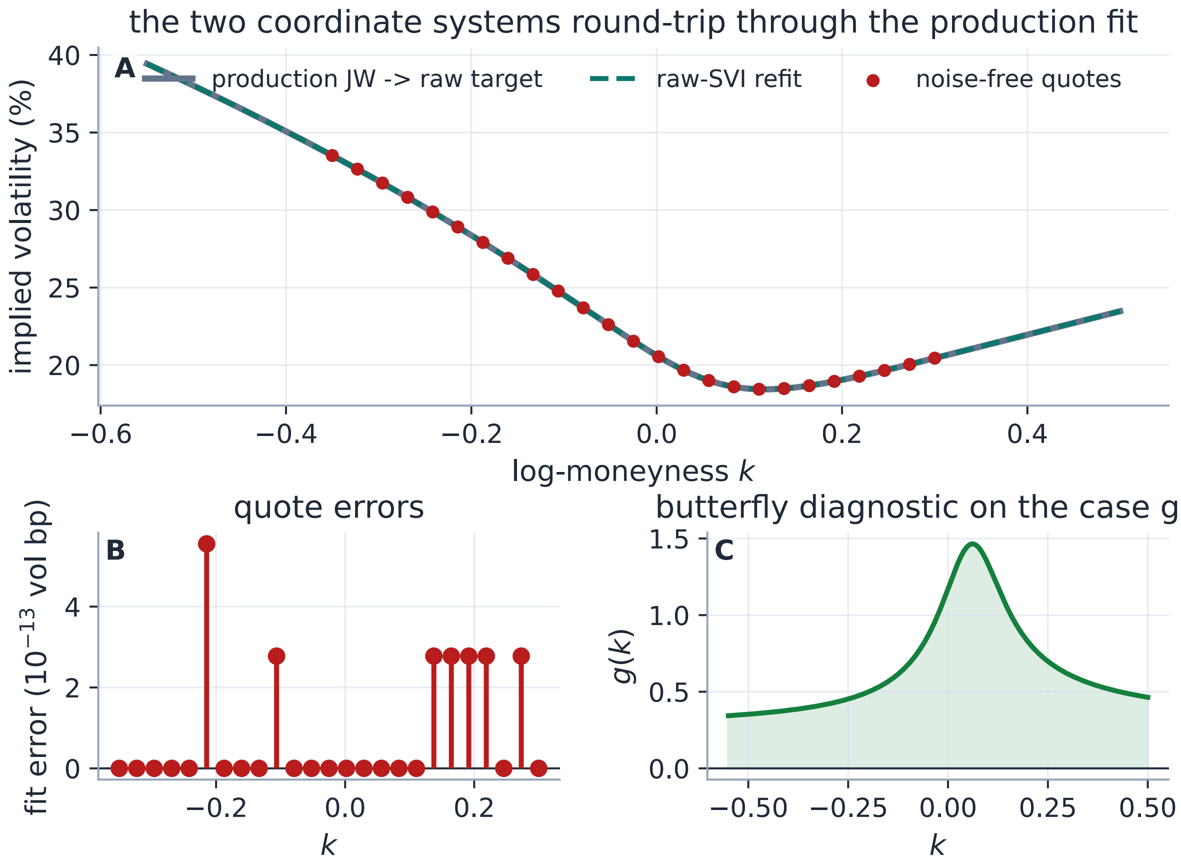

Figure 7: The laboratory target and fit coincide (A), so the useful evidence is the residual scale (B) and the independent Durrleman diagnostic (C). The test validates conversion, initialization, calibration, and reverse readout in one reproducible path; it deliberately says nothing about real-market expressiveness.

8 Where one hyperbola stops being enough

Strict convexity of total variance gives SVI one minimum and one transition between two wing slopes. This is desirable regularization on sparse equity index chains, but it is also a hard expressiveness limit. A W-shaped total-variance target, a double-humped event density, or a sharply localized short-dated kink asks the curve to turn more than once.

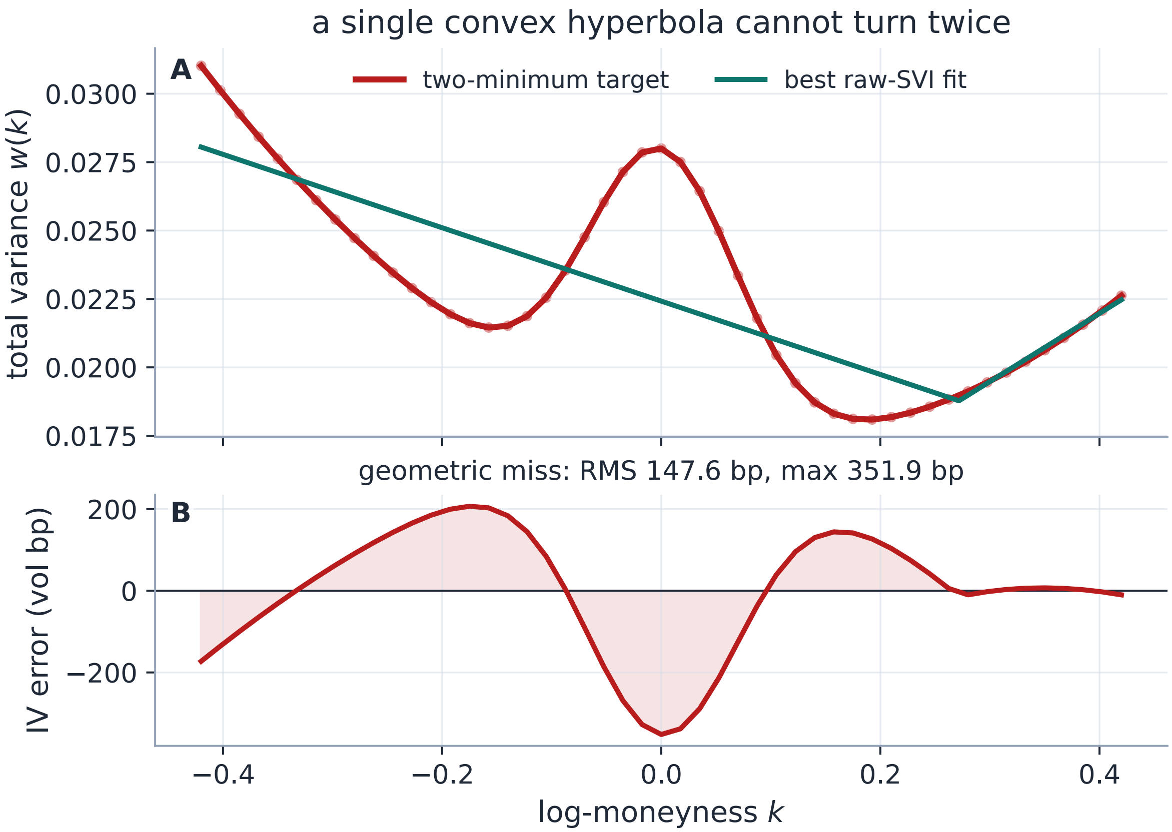

Figure 8: A deliberately synthetic positive target has two minima and a central hump. Production raw SVI returns the best one-hyperbola compromise (A), leaving an RMS miss of 147.6 vol bp and a maximum miss of 351.9 vol bp (B). This is a geometric stress test, not a claim that the synthetic target is itself an arbitrage-free market smile.

Sparse identification is a different limitation. On a narrow or one-sided quote window, distinct raw tuples can price the observed strikes almost identically while implying different remote wings. The fitted curve may be stable even when (a, b, ρ, m, s) is not. JW is easier to interpret, but its wing handles are still extrapolated functionals when the market has not quoted the wings. Across fits, compare prices, curves, and economically observed handles before comparing raw coefficients.

The historical spike-regime sweep recorded SVI at 24.3 vol bp RMS in-sample and 26.8 out-of-sample over roughly 1,576 nodes per regime. Those values are retained for continuity with the current note and deck, but the underlying result parquets are not checked into this workspace; they are historical evidence, not a result regenerated by this manuscript. The same audit changed the SVI butterfly flag rate from 20.8% under a reconstructed finite-difference diagnostic to 9.2% under the model-native analytic diagnostic. The surviving flags are consistent with the lesson of fig. 5: numerical noise exaggerated the old count, while genuine violations remain possible.

9 The honest product and mathematical contract

SVI, SVI-JW, and the no-arbitrage analysis are classical. Raw SVI is due to Gatheral; the JW chart is documented by Gatheral and Jacquier; Lee supplies the wing moment bound; Durrleman supplies the density factor; and Martini–Mingone give the full raw-SVI butterfly domain [1, 2, 3, 4, 5].

The contribution here is narrower: a clock-consistent derivation in Vol-Fitter notation, an explicit full-image theorem including the non-identified stratum, a conditioning analysis tied to the actual converter, a production-accurate separation of screens from certification, an executable checked reference map, and traceability from the mathematical claims to code and tests.

Claim or feature	Status	Qualification
raw SVI evaluation	shipped	runtime model uses `RawSVI`
raw SVI fit and storage	shipped	selected overlay is calibrated and stored in raw coordinates
five-handle JW entry/bump/export	not shipped	JW converter is used by tests and document generators, not runtime UI/API
raw $\leftrightarrow$ JW equivalence	exact	unique only on (21); singular on (22)
structural raw inequalities	hard	for finite latent coordinates in exact arithmetic
minimum and Lee controls	soft	configurable finite-weight residuals on the weak closure
global butterfly freedom	not guaranteed	requires proposition 2; counterexample is test-locked
global calendar freedom	not guaranteed	sampled finite-weight floors reduce crossings
one minimum and linear wings	structural	strength for regular smiles, limitation for event/WW shapes
analytic core Jacobian	shipped	optional var-swap/prior blocks revert to FD; extrapolation is hybrid

Table 7: The shortest reliable summary of what the model, coordinate chart, and implementation do and do not guarantee.

10 Traceability

The table names existing repository anchors. A test locks only the claim in its row: selected invalid converter cases do not constitute full input validation, and a finite diagnostic grid does not constitute a global proof.

Table 8: Mathematical and production claims with code/test anchors.

Claim	Note object	Code and test anchors

Clock-independent w; production uses working τ for annualization	eq. (2)	`api/quotes.py::PreparedQuotes`
		`api/service.py::build_slice_fit_task`
		`api/table.py`

Raw evaluation and production JW$\to$raw conversion	eqs. (3) and (19)	`models/svi_jw/svi.py` `tests/test_lqd_pricing.py::test_jw_conversion_matches_note`

Regular round trips and singular $\psi = 0$ family	theorem 1	`models/svi_jw/svi.py`
		`tests/test_svi_domain.py::test_regular_domain_round_trips`
		`::test_psi_zero_stratum_not_identified`

Claim	Note object	Code and test anchors
Selected unguarded-converter failure cases	table 2	`tests/test_svi_domain.py::test_invalid_inputs_fail_loudly_not_plausibly`
Minimum/Lee screens do not certify $g \geq 0$	fig. 5	`models/svi_jw/calibrate.py::_penalties` `tests/test_svi_domain.py::test_core_screens_do_not_certify_butterfly_freedom`

Noise-free raw recovery and curve reproduction	section 7	`models/svi_jw/calibrate.py`
		`tests/test_svi_calibrate.py::test_recovers_benchmark_parameters`
		`::test_curve_reproduced_to_machine_precision`

Analytic Jacobian and active/inactive hinge rows	section 6.3	`models/svi_jw/jacobian.py` `tests/test_svi_jacobian.py`

Band objective reaches SVI	table 4	`calib/band.py`
		`tests/test_band_fit.py`
		`tests/test_svi_jacobian.py::test_band_fit_admissible`

Sampled calendar floor and no-floor no-op	eq. (35)	`models/svi_jw/calibrate.py`
		`tests/test_overlay_calendar.py::test_svi_no_floor_is_byte_identical`
		`::test_svi_floor_crushes_calendar_violation`

Hybrid extrapolation block and advisory measurement	section 5	`calib/extrap.py`
		`tests/test_extrap_enforce.py`
		`tests/test_diagnostics.py::test_extrapolated_arb_flags_negative_g_in_wing`

Var-swap target reaches SVI	table 4	`calib/varswap.py` `tests/test_varswap.py::test_penalty_pulls_model_varswap_toward_quote_all_models`

Operator and strike-anchor priors reach SVI	table 4	`calib/operators.py`, `calib/prior.py`
		`tests/test_prior_parametric.py::test_operator_prior_pulls_all_models_toward_prior_skew`
		`::test_strike_anchor_reaches_svi_and_sigmoid`

Model-native analytic butterfly diagnostic	proposition 2	`backtest/dispatch.py::_analytic_butterfly` `tests/test_backtest_arb.py`

Settings defaults and SVI overlay wiring	appendix A	`api/schemas.py::FitSettings`
		`models/display.py::build_display_fit`
		`tests/test_api_settings.py`

A Control atlas

The table lists direct SVI controls and the shared residual controls that alter an SVI overlay. Prior persistence has its own larger atlas in Note 13; the rows here identify the entry points relevant to this calibrator.

Table 9: SVI and shared calibration controls.

Knob	Default	Role
FitSettings: direct and shared slice controls		
`model`	`lqd`	Set to `svi` to display the raw-SVI overlay.
`sviPenaltyWeight`	1000	Residual multiplier W in (32); 0 disables both core screens.
`leeSlopeMax`	2.0	Weak cap $\beta_{\max}$; values above the theoretical Lee bound are accepted by the schema.
`weightScheme`	`equal`	Unit or time-value-density quote weights. Arbitrary low-level arrays are also accepted.
`haircut`	0.005	Absolute-vol tightening used only in haircut band mode.
`midAnchorWeight`	0.05	Small mid anchor appended in bid–ask and haircut modes.
OptionsSettings / request-level shared controls		
`fitMode`	`mid`	Persisted default: mid, bid–ask band, or haircut band.
`enforceCalendar`	true	Threads the nearer displayed slice into the sampled calendar floor.
`calendarWeight`	10^6	Squared-objective weight; residual rows use its square root.
`extrapEnforce`	false	Adds tapered extrapolation rows and activates the hybrid Jacobian.
`eventsEnabled`	true	Uses event-weighted τ; when the event calendar is empty, $\tau = t$.
`varSwapEnabled`	true	Allows an available variance-swap quote to append its target row.
`varSwapWeightPct`	10	Var-swap budget as a percentage of summed option-quote weights.
`priorPersistenceMode`	`hybrid`	Routes active transported priors to strike/operator/-factor targets; absence of an active prior leaves the blocks empty.
Internal numerical choices		
$b = \mathrm{softplus}(\theta_2)$	—	Enforces $b > 0$ for finite θ_2.
$\rho = \tanh(\theta_3)$	—	Enforces $\lvert\rho\rvert < 1$ for finite θ_3.
$s = e^{\theta_5}$	—	Enforces $s > 0$ for finite θ_5.
LM tolerances	10^{-15}	`xtol`, `ftol`, and `gtol`.
trial-w floor	10^{-12}	Keeps IV residuals finite on negative-variance trial slices; not part of `RawSVI.implied_vol`.

B Performance and numerical qualifications

1. **Analytic core.** The main saving is replacing one base residual plus five parameter perturbations by one closed-form Jacobian. The generated microbenchmark is a warmed median of five timings on the current development machine; it is a regression signal, not a universal latency number.

2. **LM rather than trust region.** LM was retained after matched experiments on noisy real nodes. The result is empirical: a different residual stack or constraint strategy can change the conclusion.

3. **Conditional acceleration.** Mid/band, core penalties, and sampled calendar rows are analytic. Var-swap and prior rows trigger full finite differences. Extrapolation enforcement is hybrid.

4. **Finite arithmetic.** The hard map (33) is exact for finite real inputs. Floating-point `tanh` can round to ± 1, exponentials can underflow or overflow, and $1 - e^{-b}$ is less accurate than `-expm1(-b)` for extremely small b. The production range encountered in ordinary fits stays away from those extremes; the mathematical statement should nevertheless not be mistaken for an IEEE-754 proof.

5. **Low-level input contract.** The calibrator assumes $\tau > 0$, finite quote arrays, non-negative price-implied total variances, and enough residual rows for LM. It does not promise a meaningful five-parameter fit to an empty, tiny, or one-sided chain, nor global convergence from every start.

6. **Converter conditioning.** The checked reference listing below uses (25); it still rejects the singular stratum because no numerical rearrangement can select a unique missing curvature.

C Executable reference maps

The following file is imported and executed by `figures/gen_svi_rewrite.py`. Before figures are written, its checked inverse is compared with production `jw_to_raw` on three regular-domain cases to relative tolerance 2×10^{-12}. The inverse uses the stable denominator (25); the reverse map is exactly the functional definition (14).

Listing 1: Executable raw/JW reference maps used by this note.

```python
"""Executable reference maps for the alternative SVI-JW note."""

from __future__ import annotations

import numpy as np

from volfit.models.svi_jw.svi import RawSVI, SVIJW


def raw_to_jw(raw: RawSVI, tau: float) -> dict[str, float]:
    """Read the five JW functionals from a raw SVI slice."""
    w0 = float(raw.total_variance(0.0))
    root0 = np.sqrt(raw.m * raw.m + raw.sigma * raw.sigma)
    sqw0 = np.sqrt(w0)
    return {
```

```
        "v": w0 / tau,
        "psi": raw.b * (raw.rho - raw.m / root0) / (2.0 * sqw0),
        "p": raw.b * (1.0 - raw.rho) / sqw0,
        "c": raw.b * (1.0 + raw.rho) / sqw0,
        "v_tilde": (
            raw.a + raw.b * raw.sigma * np.sqrt(1.0 - raw.rho**2)
        ) / tau,
    }


def jw_to_raw_checked(jw: SVIJW) -> RawSVI:
    """Checked regular inverse, using a cancellation-resistant
       denominator."""
    if not (
        jw.t > 0.0 and jw.v > 0.0 and jw.p > 0.0 and jw.c > 0.0
        and -0.5 * jw.p < jw.psi < 0.5 * jw.c
        and jw.psi != 0.0 and jw.v_tilde < jw.v
    ):
        raise ValueError("JW point is outside the regular inverse
           domain")
    w0 = jw.v * jw.t
    b = 0.5 * np.sqrt(w0) * (jw.p + jw.c)
    rho = (jw.c - jw.p) / (jw.c + jw.p)
    chi = rho - 4.0 * jw.psi / (jw.p + jw.c)
    q_rho, q_chi = np.sqrt(1.0 - rho * rho), np.sqrt(1.0 - chi * chi)
    dq = (chi - rho) * (chi + rho) / (q_rho + q_chi)
    denom = ((rho - chi) ** 2 + dq**2) / (2.0 * q_chi)
    width = (w0 - jw.v_tilde * jw.t) / (b * denom)
    m = chi * width / q_chi
    a = jw.v_tilde * jw.t - b * width * q_rho
    return RawSVI(a=float(a), b=float(b), rho=float(rho),
                  m=float(m), sigma=float(width))
```

References

[1] J. Gatheral. A parsimonious arbitrage-free implied volatility parameterization with application to the valuation of volatility derivatives. Presentation, Global Derivatives, Madrid, 2004.

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[4] V. Durrleman. From implied to spot volatilities. *Finance and Stochastics*, 14(2):157–177, 2010. doi:10.1007/s00780-009-0112-1.

[5] C. Martini and A. Mingone. No arbitrage SVI. *SIAM Journal on Financial Mathematics*, 13(1):227–261, 2022. doi:10.1137/20M1351060.